Data Narrative: Decoding Workforce Dynamics - A Journey of Discovery and Insight

Introduction

In the realm of human resources, understanding the nuances of employee attrition is crucial for nurturing a dynamic and stable workforce. This project embarked on this investigative journey with a mission to offer actionable insights through the power of data visualization.

Purpose

The main aim was to explore and understand the various reasons behind employee turnover. The goal was to give HR stakeholders the tools they need to create a work environment that supports growth, commitment, and teamwork.

Audience

The visuals made for this project were specifically for HR workers who are actively involved in making and applying employee retention strategies. The success of this project relied on giving these decision-makers insights based on data.

Data Source and Credibility

The dataset that formed the backbone of this project was sourced from a Kaggle repository. Comprising 1,470 detailed employee records, it spanned across variables from job satisfaction to monthly income.
Despite its comprehensive nature, the dataset's anonymity posed questions about its credibility. This challenge was met with a balanced approach: utilizing the dataset for initial insights while acknowledging the necessity for further validation.

Chart Types and Rationale

The project utilized a variety of charts, each serving a distinct purpose:

- **Bar Charts**: These were instrumental in illustrating attrition rates, allowing comparisons across various demographics. Moreover, horizontal bar charts were also leveraged to illustrate a KPI, which gives an overall summary of the current state of employee attrition.
- **Donut Charts**: They adeptly highlighted proportions within categories, such as the distribution of attrition by gender within age cohorts.
- **Heatmaps**: Selected for their visual impact, they revealed patterns in job satisfaction across roles, prompting inquiries into potential correlations with attrition.
- **Pie Charts:** Simple pie chart was utilized to demonstrate the attrition rate across different department of the organization.

These chart types were chosen for their simplicity and effectiveness in conveying complex data to the target audience.

Spatial Chart Reflection

Facing a dataset without geographical information, I faced the challenge of how to show spatial relationships effectively. Without actual geographic data, I used DALL·E 3 AI to create a custom office layout that represents the company's departments. This was added to the Tableau dashboard not as a real map but as an interactive way to show data about different departments through unique shapes. This creative approach provided a new way to interact with the data and showed how understanding spatial relationships can improve data visualization.

Leveraging Machine Learning

To dig deeper into why employee attrition, I used the Random Forest algorithm, which helped identify the most critical factors of attrition. Despite several attempts, integrating this directly into Tableau using TabPy was challenging, so I adapted by including the findings as a bar chart on the dashboard, which highlighted the most influential factors of employee turnover, providing clear and actionable insights.

Journey and Reflection

The Inception (Feb 27, 2024)

The journey started with a keen interest in understanding why employees leave their jobs. Full of excitement but a bit unpolished, I jumped into the Kaggle dataset, which was full of valuable but unclear information. I had to be careful, making sure my eagerness to analyze didn’t overlook the need to check the data’s accuracy—a consistent theme throughout my project.

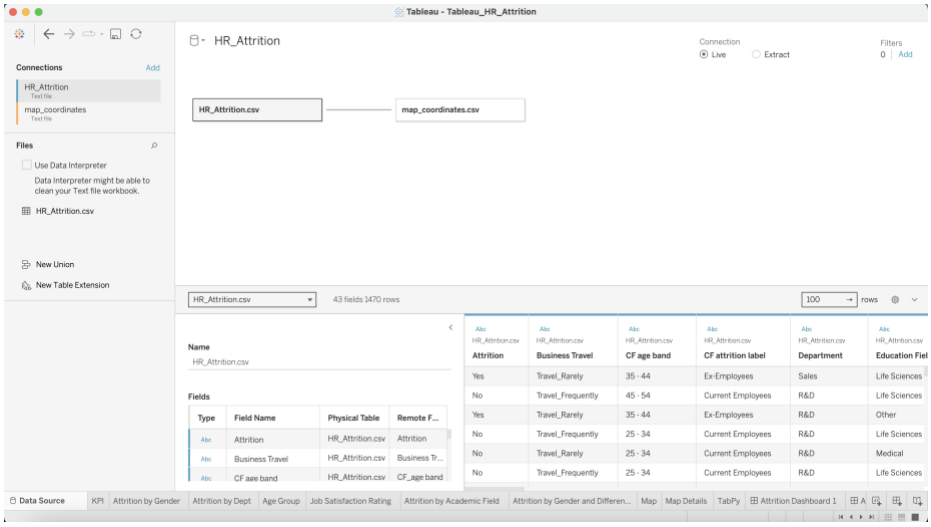

Fig 01: Importing Data into Tableau

At first, I was wondering what the attrition state was across different departments and genders. I also wanted to know the age dynamics for attrition rate.

For that I created a variable called ‘AttritionCount’ and used to an Exploratory Data Analysis for the above-mentioned aspects.

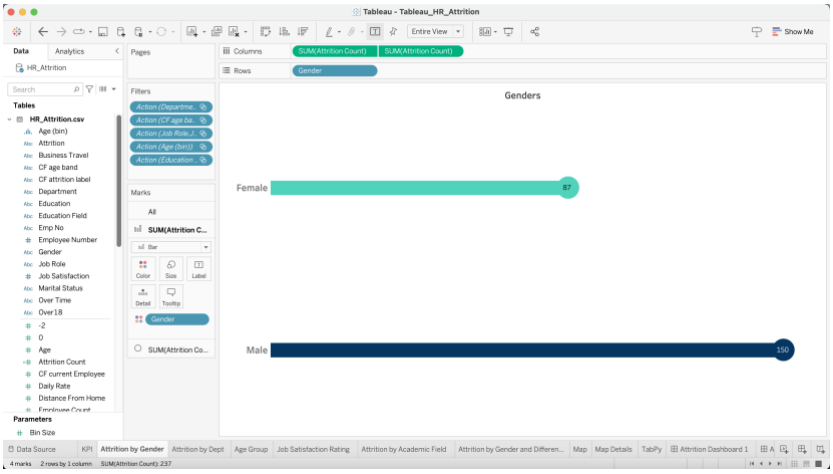

Figure 02: Attrition by Gender

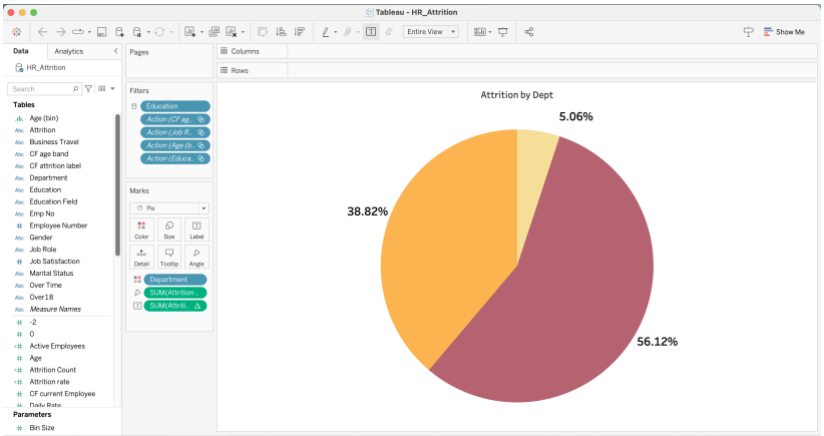

Figure 03: Attrition by Department

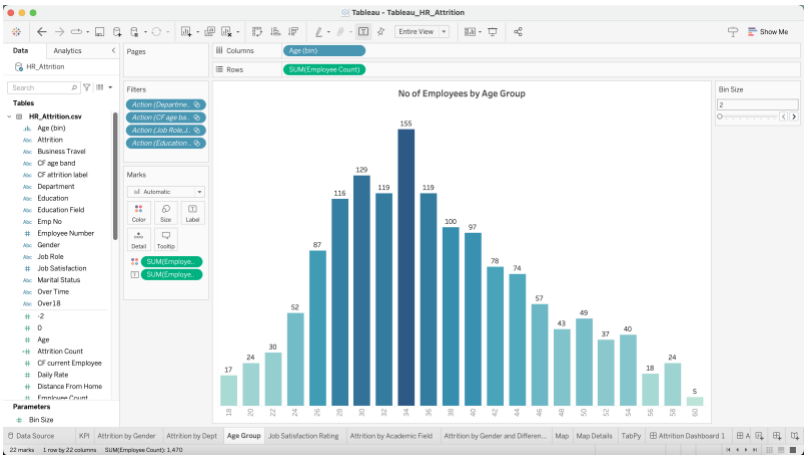

Figure 04: Attrition by Different Age Group

March 02, 2024

As I am not a very creative person, I was going through different projects on Tableau Community Projects. I saw several amazing dashboards which gave me some interesting ideas for my project. One of them was to create a KPI which gives an overall gist of the employee attrition in that organization.

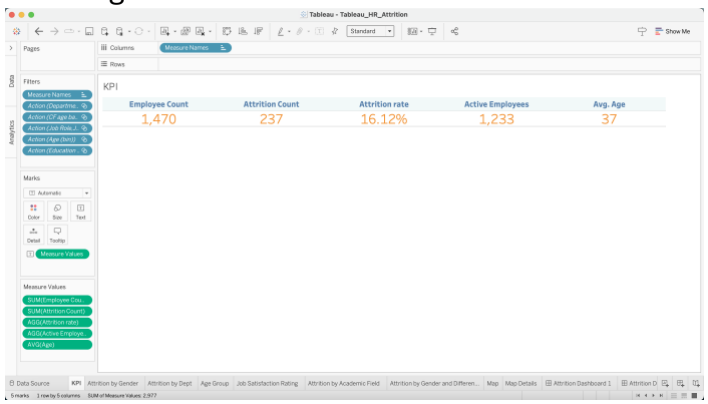

Figure 05: KPI

March 11, 2024

I saw an interesting attribute which is called 'Job Satisfaction'. Therefore I thought it would be interesting to visualize a heatmap for job satisfaction across different roles.

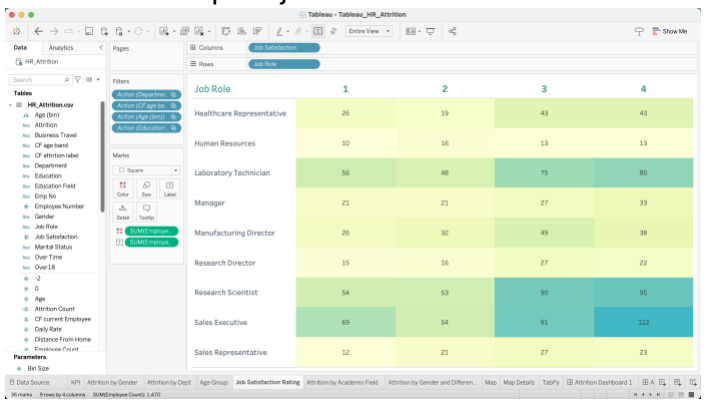

Figure 06: Job Satisfaction for Different Job Roles

Moreover, I wanted to visualize the attrition rate for employees from different academic backgrounds.

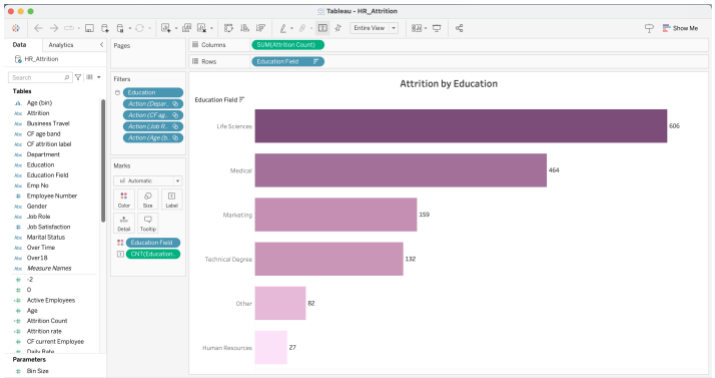

Figure 07: Attrition by Different Academic Field

Feedback Loops and Refined Insights

The project evolved with each round of feedback and critiques, like shedding light on parts of my analysis that were initially missed. Comments from peers pointed out some overlooked biases, and advice from instructors made sure I was on the right track. A critical moment was when a colleague highlighted a possible gender bias in how I was looking at the data, leading to a deeper analysis that showed how department and its respective job satisfaction played a bigger role than I first thought.

March 24, 2024

Lydia gave me this idea to visualize attrition rate for different genders for different age groups.

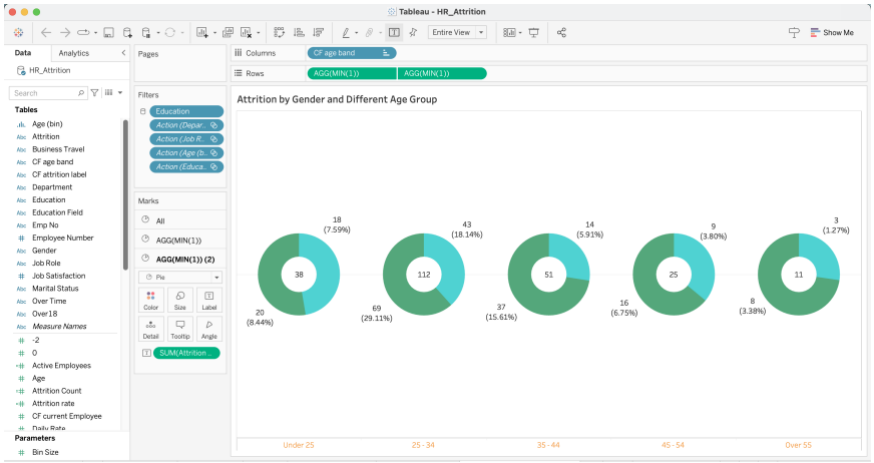

Figure 08: Attrition by Gender and Different Age Group

Due to the lack of spatial element in my data, Dr Hagman advised me to do something very interesting. Upon discussing with her, I used a custom office layout to show different visualizations for different departments of the organization.

However, I was facing issues to execute this. In the next class, Dr Hagman showed me how we can annotate different points from the image, and then create a custom coordinate to use that image as a spatial element for my dashboard.

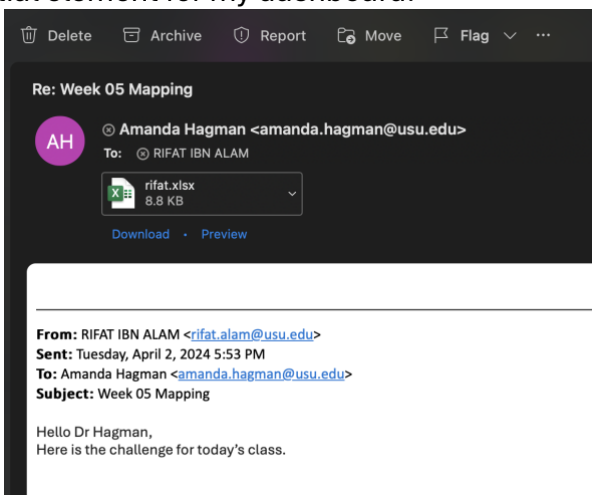

Figure 09: Email Thread with Dr Hagman

April 02, 2024

I collected an office layout from internet and tried to use this as the spatial element on my dashboard. However, I was facing with several issues on that as well.

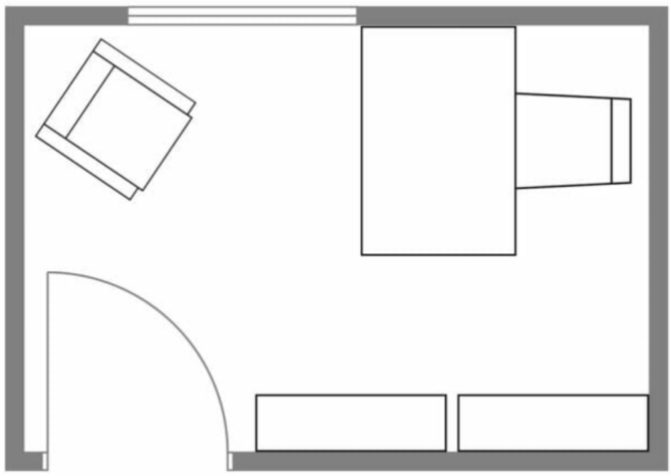

Figure 10: Office Layout (first draft)

Meanwhile, I prepared my first dashboard using some of the visualizations to continue the project. I took the suggestions from my classmates and Dr Hagman and brought several changes here. I gave an overview of Attrition into the first section as Lydia was a bit confused on what attrition is, which gave me good purpose to write a short overview on Attrition and the project itself. Also, I modified my color scheme based on the feedback of my classmates. Then, I rearranged my visualizations across 2 dashboards. Furthermore, I updated the title of my dashboard.

Figure 11: First Dashboard (before feedback)

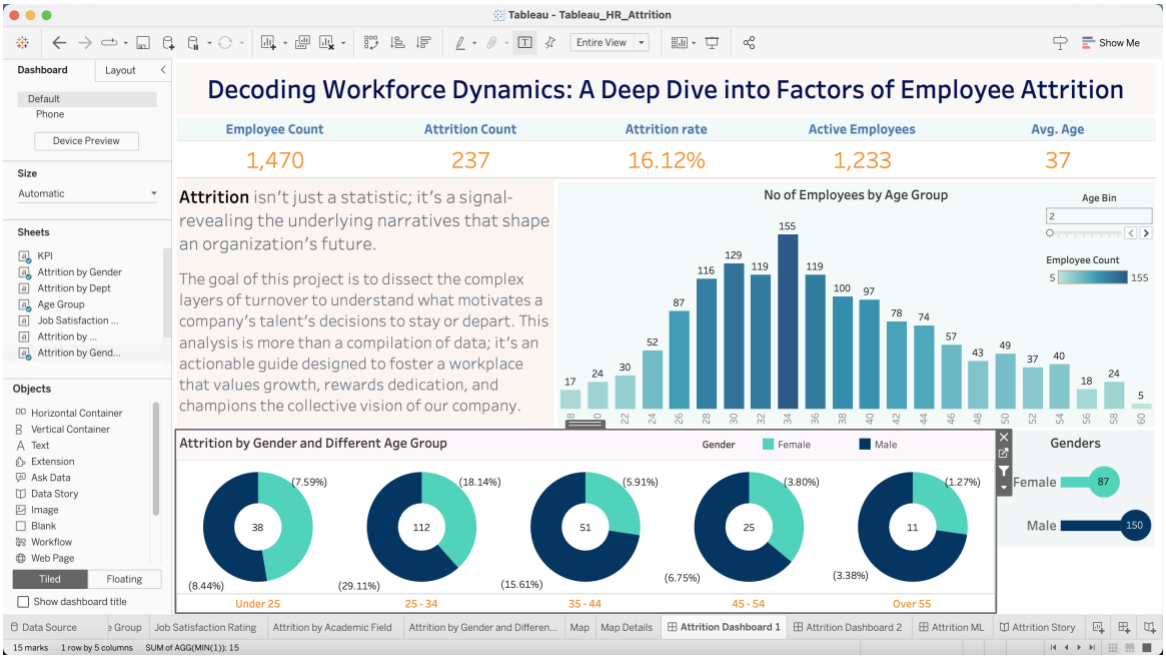

Figure 12: First Dashboard (after feedback)

April 14, 2023

I tried to integrate TabPy to apply Machine Learning in my project. However, despite several attempts, I was unable to connect the script to Tableau. Therefore, I imported the feature importance that I got using Random Forest, to my dashboard.

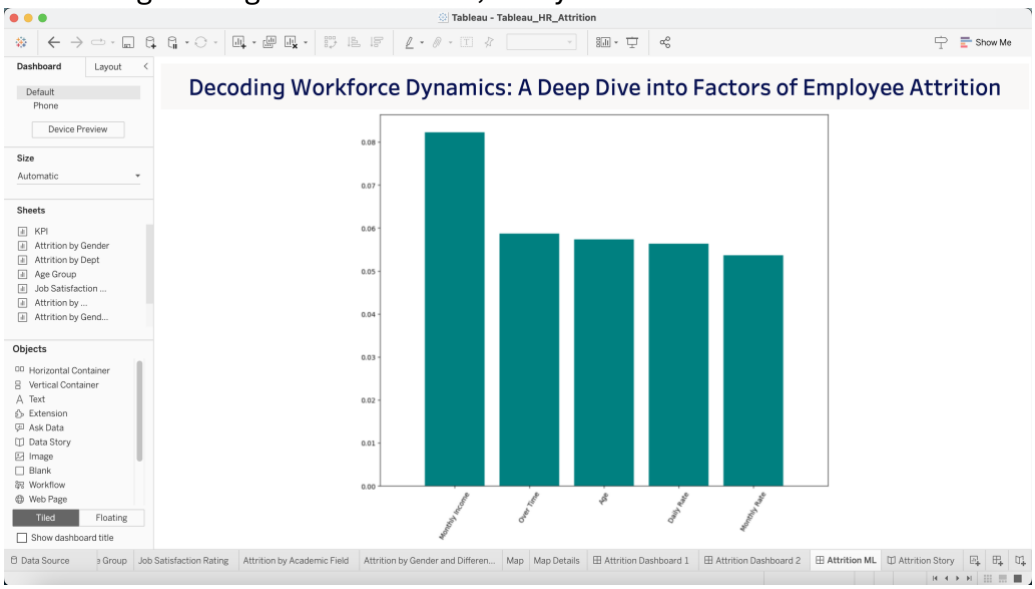

Figure 12: Feature Importance using Random Forest

April 15, 2023

I talked to Drake in his office hours. Instead of using the dull office layout, Drake gave me an amazing idea to generate office layouts using AI. We worked for a while to create the perfect image for it. However, we finalized on this, which looks much better than the previous one in my opinion.
Moreover, I was hesitant on what I should visualize using the map/ office layout. Drake and I finalized that, we can show a comparison of Employees' Monthly Income and their job satisfaction to visualize a correlation between them.

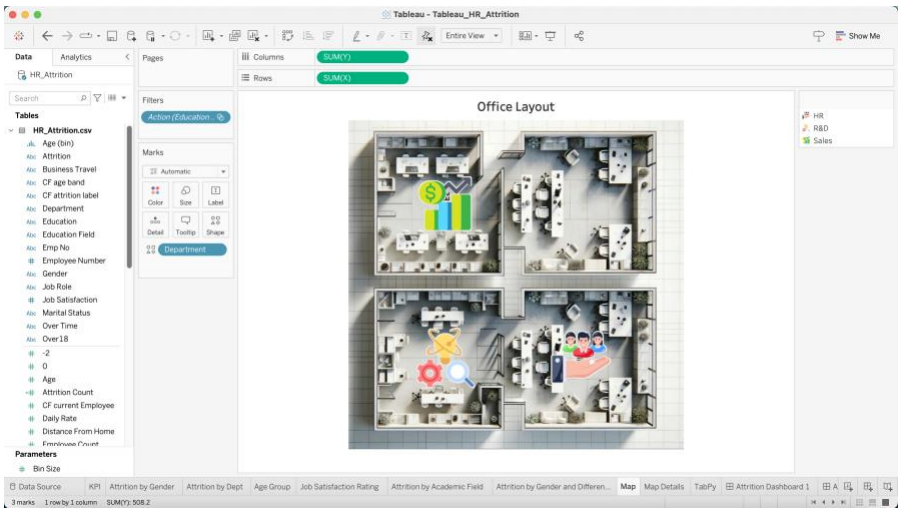

Figure 13: Augmented Office Layout

Discovery and Insights

The exploratory nature of the project shed light on the complex relationship between job satisfaction, income, and attrition, challenging some common beliefs like the direct link between satisfaction and retention. Moreover, an ambiguous result was found with the correlation between monthly income and job satisfaction. For instance, for HR professionals there was a direct correlation between their income and job satisfaction. On the contrary, employees from R&D and Sales, did not follow the same correlation.

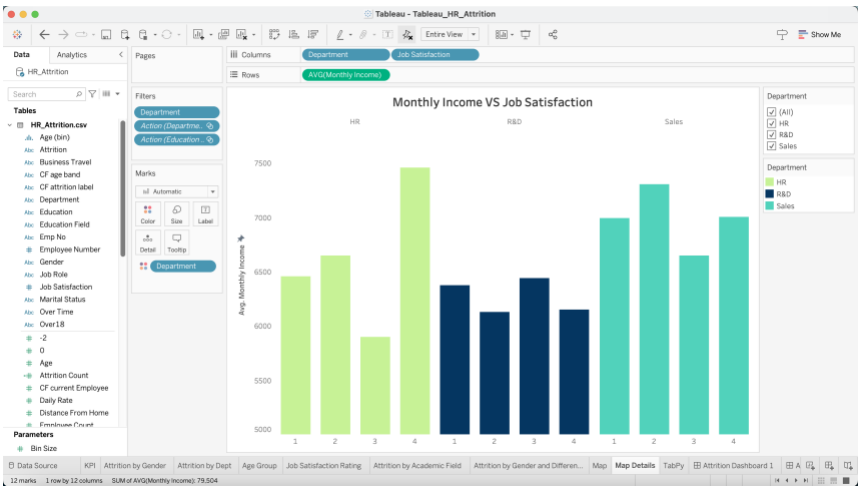

Figure 14: Monthly Income VS Job Satisfaction across Different Departments

After that I finalized my second dashboard.

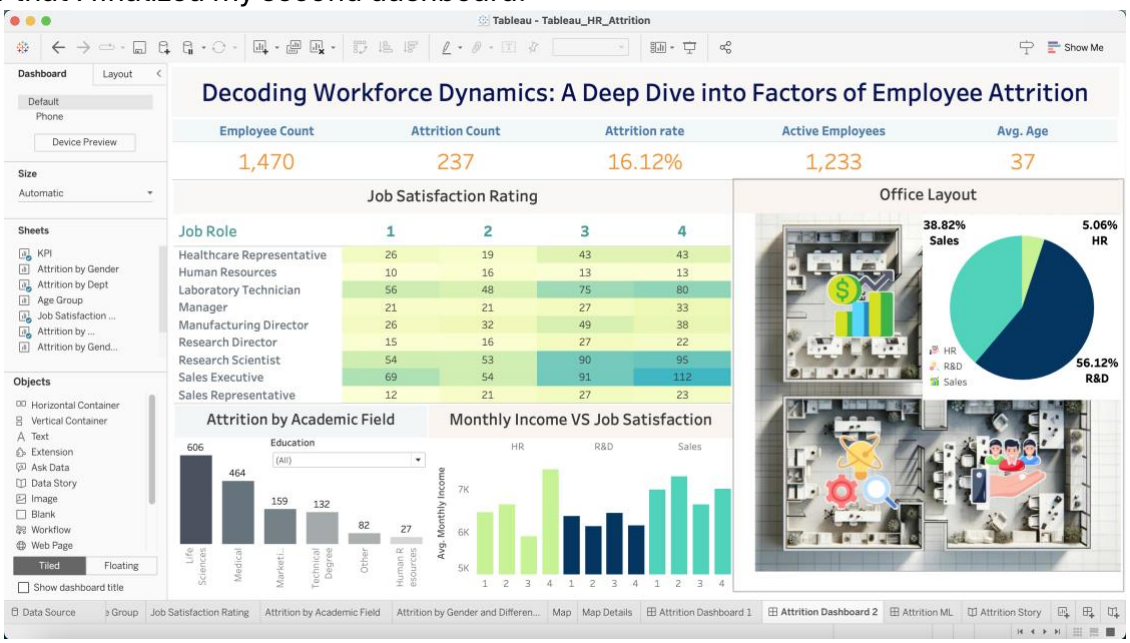

Figure 15: Second Dashboard

I followed the 'Z' pattern to create the flow of the dashboards which a very clever way to design the dashboard. Thanks for Dr Hagman for introducing this technique.
Moreover, I also put the 'Feature Importance' as the third dashboard in my project, and then combined all the dashboards into a story.

Exploratory vs. Explanatory

As the project progressed, my understanding of its nature evolved—it was exploratory in searching for new insights and explanatory in clarifying the complex realities of employee turnover, showcasing the power of data visualization not just to inform but to enlighten and initiate change.

Reflection

As I reflect on the journey from its inception to its conclusion, I see a mosaic of moments – challenges met, skills honed, and insights realized. The project was not just about numbers or charts; it was about growth and the pursuit of knowledge. It reaffirmed my belief in the transformative power of data visualization and its critical role in shaping the future of human resources and organizational development.

Conclusion

To encapsulate, this project was a profound personal and professional journey. It illuminated the potency of visual analytics as a transformative tool, translating numbers into narratives and insights into actions. It emphasized the essence of data visualization in the business domain—bringing clarity to complexity and guiding informed decision-making for the betterment of organizational health and employee well-being.